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Energy Input-Output Analyses of Major Field Vegetable Crops Production in Punjab

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ABSTRACT

The energy consumption pattern in Indian agriculture has changed with a marked shift from human and animal power to electrical and mechanical sources of energy (tractors, diesel engines, self-propelled machines). The rising cost of production and depleting energy sources are paramount concerns for the maximization of productivity with minimum input of energy and cost. A study was performed for determining the energy productivity of five major vegetable crops (potato, tomato, muskmelon, garden pea, cauliflower) production systems. Significantly higher ($p<0.05$) energy consumption was observed for potato cultivation (53,412 ± 2,388 MJ.ha^{-1}), followed by tomato (47,489 ± 1,183 MJ.ha^{-1}), cauliflower (39,367 ± 1,127 MJ.ha^{-1}), muskmelon (37,827 ± 856 MJ.ha^{-1}), and garden pea (24,625 ± 497 MJ.ha^{-1}). Irrigation was observed to be the major energy-consuming farm operation, followed by transportation of farmyard manure (FYM)/fertilisers and produce. Based on the significant contribution of non-renewable and commercial energy in the production of major vegetable crops, the agricultural system was observed to be vulnerable to increase in energy prices through petroleum as well as fertiliser prices. There is a need to include more energy- and cost-effective farm mechanisation and irrigation systems.

The amplified mandate for human livelihood via consumption of food and fibre has increased. The consumption of energy for their production has also increased. Agricultural crop production and productivity are positively correlated with energy inputs, and the best possible energy-use in agriculture has been to either boost yield with the same amount of energy input, or reduce the energy input without changing the yield (Mehta *et al.*, 2014; Jat *et al.*, 2020). Implementing extreme yielding varieties, exhaustive cropping approaches, extending irrigation conveniences, using more fertilisers and chemicals, and having a higher level of mechanisation have dramatically increased agricultural production. Consequently, the structure of energy consumption has reformed with a shift from animal and human power to tractors, electricity, and diesel power (Sharma *et al.*, 2014; Raghuvanshi *et al.*, 2018). The expenditure configuration of energy inputs scrutinized by other researchers indicated that the energy consumption per hectare and the gross cropped area increased, and consequently the output per unit of energy-use decreased (Mehta *et al.*, 2014).

Animal and human power, the two major renewable energy sources, were comprehensively and traditionally used for crop production in Punjab as well as in Indian agriculture. Before 1960–61, draft animals were the foremost source of motive power (tractive and rotary), and to a larger extent for agricultural crop production, water lifting, rural transport, oil extraction, sugarcane crushing, chaff cutting, and transportation (Lohan *et al.*, 2015). The energy contributed by draft animals in

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crop production in Punjab has decreased from 9,785 MJ.ha^{-1} to 2,054 MJ.ha^{-1} between 1970–71 and 2015–16 (Table 1). In contrast, with increase in population, the population density (605 - 731 per 1,000 ha) and energy contribution (2,846 - 4,013 MJ.ha^{-1}) from agricultural workers have increased (Lohan *et al.,* 2015; Anon., 2020).

The non-renewable energy-uses by tractor, farm machinery, stationary engine, electricity, fertiliser, and agro-chemical have increased during the last five decades. Non-renewable energy sources are prerequisites for all farm operations from tillage to harvesting, and also for transportation. In addition to the increase in number of tractors (39,800 to 452,000) and diesel engines (7,000 to 194,000), the global utilisation of diesel in agriculture sector has increased from 274,750 (3817 MJ.ha^{-1}) to 24,74,660 thousand litres (33,740 MJ.ha^{-1}) between the year 1970 and 2020 (Fig. 1). As the number of power-driven tube wells being operated on diesel engine / electric motor increased from 192,000 to 935,000 between 1970–71 and 2015-16 (Lohan *et al*., 2015; Anon., 2020), the consumption of electricity in agriculture has also

Table 1. Time series annual amount of energy inputs* for agriculture crop production in Punjab

SI. No.	Year	Agricultural Worker, '000	Draught Animal, '000	Diesel consumption, 1000 l	Consumption of Electricity in agriculture, 1000 kW.h	Total chemicals (Pesticides/ Insecticides), 1000 kg	Total Fertiliser (N, P, K), 1000 kg
1.	1970-71	2,452	1,636	2,74,750	463	-	213
2.	1975-76	3,218	1,670	5,61,500	1,063	3,300	295
3.	1980-81	2,860	1,649	8,34,728	1,850	4,480	762
4.	1985-86	3,115	1,649	10,39,850	2,799	6,400	1,098
5.	1990-91	3,370	1,585	12,18,790	5,105	9,100	1,220
6.	1995-96	3,463	1,521	19,01,270	5,735	10,656	1,263
7.	2000-01	3,555	436	22,78,755	5,534	10,640	1,313
8.	2005-06	3,647	436	23,78,140	7,314	8,465	1,687
9.	2010-11	3,555	382	24,58,483	10,117	8,227	1,911
10.	2015-16	3,523	350	24,74,660	11,514	8,094	1,943

Source: Singh, 2012; Anon., 2020

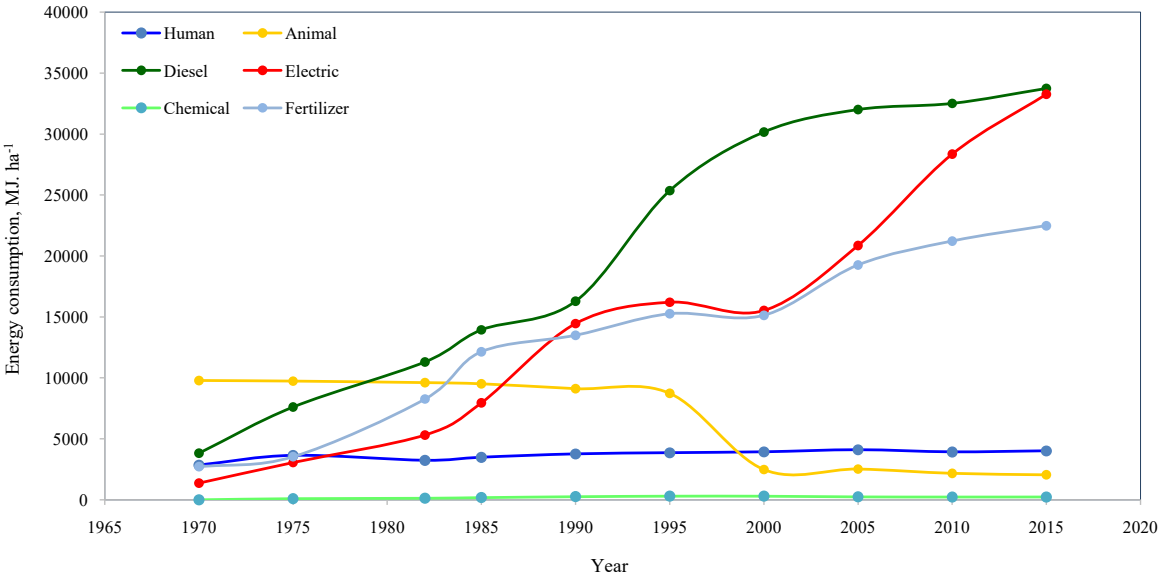

Fig. 1: Time series source-wise energy consumption for agriculture crop production in Punjab

(*Energy consumption was calculated by multiplying the number of inputs obtained from Table 1 to their energy equivalents divided by net cropped area of Punjab state; *(Source: Singh, 2012; Anon., 2020)*

increased from 1,363 MJ.ha^{-1} to 33,260 MJ.ha^{-1} during the same period.

Fertilisers and agro-chemicals are exceptionally energy-demanding commodities, and represent almost 50% of total trade energy input (Singh, 2012). The consumption of fertilisers increased from 213 thousand kg in 1970–71 to 1,943 thousand kg in 2015–16. The energy supplied per ha through fertilisers and agro-chemicals increased from 2,713 MJ.ha^{-1} to 22,482 MJ.ha^{-1}, and from 95 MJ.ha^{-1} to 235 MJ.ha^{-1}, respectively. Future agricultural strategies would attempt to reduce energy inputs by employing such replacements as crop rotations and green manures to lessen the elevated energy need of chemical fertilisers and pesticides.

The agricultural production scenario, particularly in Punjab, has massively transformed since the last few epochs. Out of the total 5.03 Mha geographical area, the net sown area in Punjab has increased from 4.05 to 4.13 Mha between 1970–71 and 2015–16 (Lohan *et al.,* 2015; Anon., 2020). Presently, about 0.273 Mha is under vegetable crops cultivation with an aggregate production of 5.44 Mt (Kumar and Kaur, 2018; Anon., 2020). Potato, pea, cucurbits, cauliflower, and tomato are the predominantly grown vegetable crops in the state. Amid these, about half of the net sown area increase is contributed to potato crop, which is cultivated as fresh market produce and seed potato. Regardless of the magnitude of vegetable crops and their increasing acreage, there is minimal information on their energy usage.

Consequently, there is a prerequisite to investigate the energy-use of vegetable production systems (both source and farm operation) to determine the effect of energy input on vegetable crop productivity, and to assess optimum energy-use levels. Further, crop-energy analysis needs to be performed to evaluate the energy requirements of the crops, and endorse non-conventional renewable sources of energy involving technologies, institutions, and policy measures. The major objective of this research was to investigate the energy-use patterns and to analyse energy input–output in the cultivation of major vegetable crops in Punjab.

MATERIAL AND METHODS

The study was conducted in the College of Agricultural Engineering and Technology, Punjab Agricultural University (PAU), Ludhiana with the collaboration of PAU Vegetable Research Farm, Khanaura, Hoshiarpur, and PAU Krishi Vigyan Kendra, Hoshiarpur.

Field data was compiled during the period of two complete consecutive *rabi* and *kharif* crop seasons for the year 2020-2021 and 2021-22 after due consideration of statistical techniques. These years were characterised by normal weather in Punjab without any extreme weather events which could have an impact on the crop production (Anon., 2021; 2022; 2023).

Study Area

Amongst the total five agro-climatic zones and 21 districts of Punjab state, three districts (Hoshiarpur, Jalandhar, SBS Nagar), the major vegetable crop producing districts, were selected.

A total of 12 villages from the above three districts were selected for the study. Majority of the vegetable growing farms were of marginal and small land holdings. Therefore, 10 farmers in each village producing five major vegetable crops, viz. potato, muskmelon, tomato, garden pea, and cauliflower were selected for the survey. The selected three districts had large share of marginal (<1 ha) and small (1-2 ha) farmers (Anon., 2020). Hence, marginal, and small category of farmers were selected for the survey to maintain uniformity as the farm sizes do not vary considerably.

Data Collection

The data collection involved two questionnaires; i) Performa part-I for Information regarding operational inputs/outputs from various energy sources, and ii) Performa part-II for specifications and performance data of farm machinery used for the cultivation.

The operational inputs were collected through personal interactions with the farmers at different stages of crop production (nursery raising, transplanting, intercultural operations, multiple harvestings) for each crop. The information pertaining to different sources of inputs, specifically human, fuel, farm machines, seed, FYM, fertilisers, and chemicals for carrying out various farm operations were collected and analysed. Summarized data on the amount, replications, and method of various inputs are presented in Table 2. Major specifications of farm machines used by the farmers are illustrated in Table 3.

Table 2. Application methods and quantity of inputs, number of various operations employed for production of selected vegetable crops in study area

SI. No.	Farm operation / Inputs / methods	Potato	Tomato	Muskmelon	Garden pea	Cauliflower
1.	*Seed drying/ treatment*	Manual	-	-	Manual	-
2.	Human labour engaged, man.h.ha^{-1}	90-104	-	-	3-4	-
3.	*Nursery raising*	-	Manual	Manual	-	Manual
4.	Human labour engaged, man.h.ha^{-1}	-	67-80	67-80	-	67-80
5.	*Tillage operation*	Disc harrow (2), Cultivator (1), Planker (1), Laser Leveller (1)	Disc harrow (2), Cultivator (1), Planker (1), Laser leveller, (1), Bed former (1)	Disc harrow (2), Cultivator (1), Planker (1), Laser leveller (1), Bed former (1)	Disc harrow (2), Cultivator (1), Planker (1), Laser leveller (1)	Disc harrow (2), Cultivator (1), Planker (1), Laser leveller (1), Bed former (1)
6.	Human labour engaged, man.h.ha^{-1}	9-14	9-14	9-14	9-14	9-14
7.	*Sowing/ planting/ transplanting operation*					
8.	Crop variety	Kufriphukraj	Hybrids	Hybrids	AP-3	Hybrids
9.	Seed rate, kg.ha^{-1}	3250.0-4500.0	0.250-0.300	1.0	75.0 - 112.5	0.625-1.250
10.	Method of sowing/ planting/ transplanting	Automatic/ semi-automatic potato planter	Manual transplanting	Manual transplanting	Ridger seeder/ Pea seed drill-cum-bed former	Manual transplanting
11.	Human labour engaged, man.h.ha^{-1}	20-30	67-83	67-83	3-4	67-83
12.	*Bund making*					
13.	Machine used	Bund former (1)	Bund former (1)	Bund former (1)	Bund former (1)	Bund former (1)
14.	Human labour engaged, man.h.ha^{-1}	1	1	1	1	1
15.	*Transportation and application of fertilisers*					
16.	Transportation of fertilisers	Tractor drawn trailer (1)	Tractor drawn trailer (1)	Tractor drawn trailer (1)	Tractor drawn trailer (1)	Tractor drawn trailer (1)
17.	Application of fertilisers	Manual broadcasting (1)	Manual broadcasting (1)	Manual broadcasting (1)	Manual broadcasting (1)	Manual broadcasting (1)
18.	Human labour engaged, man.h.ha^{-1}	9-11	9-10	9-10	8-10	9-10
19.	FYM, kg.ha^{-1}	25000-30000	25000-30000	24000-26000	20000-21000	25000-30000
20.	Nitrogen, kg.ha^{-1}	187.50-200	150-155	120-125	50-75	125-140
21.	Phosphorus, kg.ha^{-1}	62.5-65	60-65	60-65	62.5-75	62.5-65
22.	Potassium, kg.ha^{-1}	62.5-65	60-65	60-65	0-10	56.5-65
23.	*Intercultural operation*					
24.	Weeding	Wheel hand hoe (1)	Wheel hand hoe (1)	Wheel hand hoe (1)	Wheel hand hoe (0-1)	Wheel hand hoe (4)
25.	Earthing up	Ridger (1)				
26.	Human labour engaged, man.h.ha^{-1}	17-19	17-20	15-18	0-17	65-80
27.	*Irrigation*					
28.	Method of irrigation	Manual spade, submersible pump (4-5)	Manual spade, submersible pump (11-13)	Manual spade, submersible pump (9-10)	Manual spade, submersible pump (2-3)	Manual spade, submersible pump (9-10)
29.	Human labour engaged, man.h.ha^{-1}	44-50	120-130	100-111	22-27	90-120
30.	*Spraying operation*					
31.	Method of spray for weed control	Tractor-operated boom sprayer (1)	Manual knapsack sprayer (1)	Nil	Manual knapsack sprayer (1)	Nil
32.	Method of spray for disease control	Tractor-operated boom sprayer (2-3)	Manual knapsack sprayer (1-2)	Manual knapsack sprayer (0-1)	Manual knapsack sprayer (1-2)	Manual knapsack sprayer (1-2)
33.	Method of spray for insect control	Tractor-operated boom sprayer (1)	Manual knapsack sprayer (3-4)	Manual knapsack sprayer (3-4)	Manual knapsack sprayer (0-1)	Manual knapsack sprayer (2-3)

(contd.)

(contd. Table 2)

34.	Human labour engaged, man.h.ha^{-1}	3	63-70	38-50	25-40	38-45
35.	Total chemicals/ pesticides used, kg.ha^{-1}	6.25- 8.25	7.50-9.50	4.0-6.0	4.50-5.50	4.0-7.0
36.	*Harvesting/ digging/ plucking/collection*					
37.	Method of harvesting	Digging by potato digger (1), Manual collection (1)	Manual (5)	Manual (2)	Manual (2-3)	Manual (2)
38.	Human labour engaged, man.h.ha^{-1}	234-250	1110-1430	444-667	667-750	488-500
39.	Transportation of produce					
40.	Method of transportation	Loading/ unloading (manual), Transportation of produce by tractor drawn trailer (1)	Loading/ unloading (manual), Transportation of produce by tractor drawn trailer (5)	Loading/ unloading (manual), Transportation of produce by tractor drawn trailer (2)	Loading/ unloading (manual), Transportation of produce by tractor drawn trailer railer (2-3)	Loading/ unloading (manual), Transportation of produce by tractor drawn trailer (2)
41.	Human labour engaged, man.h.ha^{-1}	8-15	40-73	16-29	24-43	16-28
42.	*Yield, kg.ha^{-1}*	28000-30000	60000-62000	28000-30000	7000-7500	34000-35000

Note: Figures in parentheses denote number of operations performed

Table 3. Major Specifications of farm machines for calculation of input energy

SI. No.	Farm operation	Machinery used	Average power/ size/ working width of machine	Effective field capacity of machine, ha.h^{-1}	Weight of machine, kg	Expected working life of machine, h
1.	Tillage operation	Tractor	33.55-41.01 kW	*	2000	10000
		Disc harrow	2.60 m	0.65-0.72	450	3000
		Cultivator	2.60 m	0.66-0.77	150	4000
		Planker	2.60 m	0.85-0.125	100	1500
		Laser leveler	2.10 m	0.10-0.15	225	2000
		Bed former	2.10 m	0.65-0.75	100	1500
2.	Sowing/planting	Automatic/ semiautomatic potato planter	1.20 m	0.10-0.15	200	2000
		Pea seed cum fertiliser drill	2.10 m	0.45-0.55	400	2500
3.	Bund making	Bund maker	1.20 m	1.85-2.15	110	1500
4.	Transportation and application of FYM/ Fertiliser	Tractor trailer	2.2 x 4.3 m	0.225-0.275	1500	3600
5.	Intercultural operation	Weeding (wheel hand hoe)	0.15 m	0.11-0.125	8	1500
		Earthing up (ridger)	1.80 m	0.65-0.75	100	1500
6.	Irrigation	Manual spade	0.10 m	0.083-0.10	2.1	1000
		Submersible pump	9.32 kW	0.083-0.10	100	10000
7.	Spraying operation	Tractor operated boom sprayer	9.20 m	1.50-1.75	350	2000
		Manual knapsack sprayer	1.00 m	0.08-0.10	10	2000
8.	Harvesting/ digging	Potato digger	1.20 m	0.275-0.35	300	2500
9.	Transportation of produce	Transportation of produce (tractor and tractor trailer)	2.2 x 4.3 m	0.225-0.275	1500	3600

**The effective field capacity and man.h.ha^{-1} of tractor depends upon the use of concerned farm machine attached to it*

Norms and Calculations of Energy Input and Output Analysis

The following norms and procedures were followed for input/output energy analyses:

(i) Only accountable forms of materials and energy inputs to a production system were considered as inputs of the system, which were consumed for the production, processing, transport, distribution, storage, and maintenance of materials or services during its useful life (Panesar and Bhatnagar, 1994).

(ii) The energy norm of machinery or materials was uniformly distributed over the total life span of the machinery, Recoverable forms of produce and energy outputs were determined as outputs of the system. As the residues/biomass of crop production system left in the field were not recoverable, they were not considered for determining output energy (Panesar and Bhatnagar, 1994).

Energy Source Classifications

The energy obtained from various sources was analysed based on crop, source, and farm operation. The energy sources were classified as renewable / non-renewable and direct / indirect energy sources. The renewable and non-renewable sources are based on high and low rates of its replenishment. The inputs having higher rate of replenishment such as human labour, draught animal, farm yard manure (FYM), crop residue, and seed are termed as renewable energy inputs (Jat *et al.,* 2020). Other energy inputs with lower rate of recharge, such as petroleum products, electricity, fertilisers, chemicals, and machinery are termed as non-renewable energy. Non-renewable energy sources were further sub-grouped depending on their source of production, e.g., hydroelectric, or thermal (e.g., diesel, coal, nuclear). Furthermore, the direct sources of energy (human, diesel, electricity) release energy directly, whereas the indirect sources of energy (machinery, seed, fertilisers, chemicals) release energy through conversion process.

The commercial energy sources involved in crop production are seed, diesel, electricity, chemicals, fertilisers, and machinery; whereas, non-commercial energy sources include human, and FYM. The classifications of energy with their sources of energy are presented in Fig. 2. The energy equivalents of various inputs and outputs in vegetable production are presented in Table 4. Hybrid vegetable seeds were most popular among the farmers. The seed rates of the crops depend on the kind of planting material. For instance, potato is vegetatively propagated, and thus requires higher seed rate. On the other hand, transplanted crops as tomato, muskmelon, and cauliflower require very low seed rates for nursery planting. Row-to-row and plant-to-plant spacing vary according to the growth habit and canopy of different cultivars and plant species.

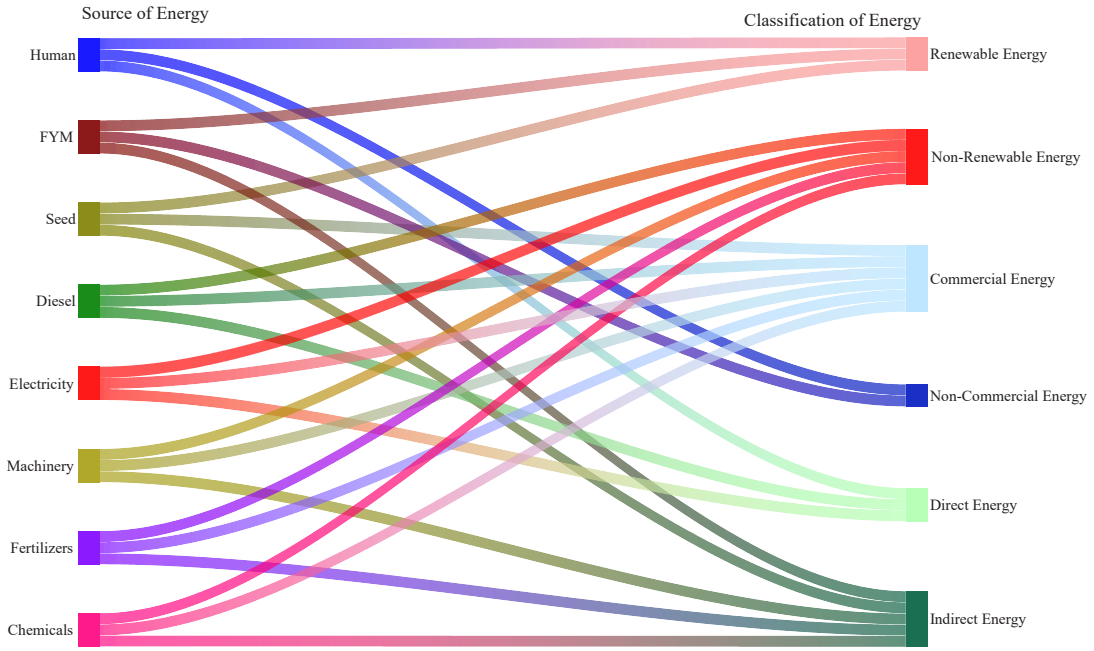

Fig. 2: Farm energy sources grouped under different categories of energy

Table 4. Energy equivalents of various inputs and outputs in vegetable crops production

SI. No.	Energy source	Energy equivalent	Unit
1.	Human	1.96	$MJ.h^{-1}$
2.	Animal	10.1	$MJ.h^{-1}$
3.	Prime mover/ tractor	68.4	$MJ.kg^{-1}$
4.	Self-propelled machine	68.4	$MJ.kg^{-1}$
5.	Farm machinery	62.7	$MJ.kg^{-1}$
6.	Electric motor	64.8	$MJ.kg^{-1}$
7.	Diesel (fuel energy)	56.31	$MJ.l^{-1}$
8.	Electricity	11.93	$MJ\ (kWh)^{-1}$
9.	Seed		
	a) Tuber crop (potato)	3.6	$MJ.kg^{-1}$
	b) Fruit / seed vegetable (tomato, garden pea)	1.9	$MJ.kg^{-1}$
	c) Cucurbitaceous (muskmelon)	0.8	$MJ.kg^{-1}$
	d) Cole crop (cauliflower)	1.2	$MJ.kg^{-1}$
10.	Fertiliser		
	a) Farm yard manure (FYM)	0.3	$MJ.kg^{-1}$
	b) Nitrogen (N)	60.6	$MJ.kg^{-1}$
	c) Phosphorus (P)	11.1	$MJ.kg^{-1}$
	d) Potassium (K)	6.7	$MJ.kg^{-1}$
11.	Chemical	120.0	$MJ.kg^{-1}$
	a) Insecticide (Imidaclorprid, Deltamethrin), Fungicide (Di-M-45, Ridomil), weedicide (Pendimethalin/ stomp)		

Source: Panesar and Bhatnagar, 1994; Jat et al., 2020

Among the selected five vegetable crops, garden pea as a leguminous crop required lowest nitrogen requirement. In contrast, potato crop required highest quantity of nitrogen as it tends to attain higher plant mass during a short period while compensating for the adequate growth of shoots as well as tubers.

The energy consumed from various sources viz., human, fuel, machinery, electricity, seed, FYM, fertilisers, and chemicals were determined as following (Jat *et al.,* 2020):

Human energy input ($MJ.ha^{-1}$) for each filed operation (tillage, sowing/transplanting/intercultural/ spraying/ fertiliser/irrigation/harvesting, transportation) was based on the actual $man.h.ha^{-1}$ used for the operation and human energy equivalents.

$$H_e = \frac{\text{No.of human labour used x time consumed (h)}}{\text{Area covered (ha)}} \text{ x Human energy equivalent } (MJ.h^{-1}) \quad \ldots(1)$$

The energy contribution of machinery for each farm operation was determined based on the weight of machine, its estimated wear-out life, effective field capacity, and energy equivalents of a machine.

$$M_e = \frac{\text{Weight of specific machine (kg) x Energy equivalent of machinery } (MJ.kg^{-1})}{\text{Wear out life of the machine (h) x Effective field capacity } (ha.h^{-1})} \quad \ldots(2)$$

Fuel (diesel) consumption for operating tractor/ engine-operated farm machines to do various farm operations was based on the actual fuel consumed, effective field capacity of the machine, and diesel energy equivalent.

$$D_e = \frac{\text{Fuel consumption } (l.h^{-1})}{\text{Effective field capacity } (ha.h^{-1})} \text{ x Fuel energy equivalent } (MJ.\ l^{-1}) \quad \ldots(3)$$

Electric energy for pumping water was calculated based on the amount of electricity consumed, irrigation capacity of pump, and the energy equivalents of electricity.

$$E_e = \frac{\text{Electricity consumption } (kWh\ .h^{-1})}{\text{Irrigating capacity of the pump } (ha.h^{-1})} \text{ x Electricity energy equivalent, } MJ(kW.h)^{-1} \quad \ldots(4)$$

The energy contributed by other sources of inputs (seed, FYM, fertiliser, chemical) were calculated by the amount of the respective inputs applied per unit of area, and energy equivalents of the source of input.

$$E_{is} = \text{Amount of applied inputs}(\text{kg.ha}^{-1}) \times \text{Input energy equivalent}(\text{MJ.kg}^{-1}) \quad \ldots(5)$$

Where,

H_e = Human energy used, MJ.ha^{-1},
M_e = Energy used from machinery/tractor, MJ.ha^{-1},
D_e = Energy used from diesel fuel, MJ.ha^{-1},
E_e = Energy used from electricity, MJ.ha^{-1}, and
E_{is} = Energy used from seed, FYM, fertilisers, chemicals, MJ.ha^{-1}.

Energy Analysis

The following parameters were considered to analyse the input-output energy for the production of vegetable crops:

Total input energy (source), MJ.ha^{-1} = Energy contributed by (Human + Diesel + Electricity + Machinery + Seed + FYM + Fertilisers+ Chemicals), MJ.ha^{-1} …(6)

Total input energy (farm operation), MJ.ha^{-1} = Energy consumed in (Seed drying/treatment + Nursery raising + Tillage operation + Sowing/planting/transplanting + Bund making + Intercultural operation + Transportation and application of FYM/ Fertiliser + Spraying operation + Irrigation + Harvesting/digging/plucking + Transportation of produce including loading/unloading + Inputs for crop management), MJ.ha^{-1} …(7)

$$\text{Output energy, MJ.ha}^{-1} = \text{Crop yield (kg.ha}^{-1}) \times \text{Energy equivalent of seed (MJ.kg}^{-1}) \quad \ldots(8)$$

$$\text{Net energy (MJ.ha}^{-1}) = \text{Output energy (MJ.ha}^{-1}) - \text{Input energy (MJ.ha}^{-1}) \quad \ldots(9)$$

$$\text{Energy} - \text{use efficiency (EUE)} = \frac{\text{Output energy (MJ.ha}^{-1})}{\text{Total energy input (MJ.ha}^{-1})} \quad \ldots(10)$$

$$\text{Specific energy consumption, (MJ.kg}^{-1}) = \frac{\text{Total energy input (MJ.ha}^{-1})}{\text{Seed yield (Kg ha}^{-1})} \quad \ldots(11)$$

$$\text{Energy productivity, (kg.MJ}^{-1}) = \frac{\text{Seed yield (kg.ha}^{-1})}{\text{Total energy input (MJ.ha}^{-1})} \quad \ldots(12)$$

Statistical Analysis

Data corresponding to energy input of selected vegetable crops were analysed using SPSS software (24.0 version).

One-way ANOVA, followed by post-hoc comparison using Tukey's honestly significant difference (HSD) test were applied to identify significant differences among energy inputs and crops.

RESULTS AND DISCUSSION

The energy-use patterns in the production of selected vegetable crops are reported in terms of human, seed, FYM, diesel, electricity, machinery, fertiliser, and chemical energy. In the present study, input energy consumed in production of the vegetable crops are reported on source and farm operation basis.

Consumption of Energy in Vegetable Production

Energy sources

The consumption of various energy sources on vegetable cultivation were classified under renewable/ non-renewable, direct/indirect, and commercial/non-commercial sources (Fig. 3).

It was observed that the average contributions of renewable sources (human, seed, FYM) were in the range of 25.57% to 43.57% of total input energy for production of the selected vegetable crops. In contrast, non-renewable sources of energy (diesel, electricity, machinery, fertilisers, chemicals) contributed 56.43% to 74.43% of the total energy input. The direct (human, diesel, electricity) and indirect (seed, FYM, fertiliser, chemical, machinery) sources of energy contributed 54 - 54.04% and 47.85 - 69.46%, respectively, of total energy consumption. The commercial (diesel, electricity, seed, fertiliser, chemical, machinery) and non-commercial (human, FYM) sources of energy contributed 68.51 - 82.82% and 17.18 - 31.5%, respectively, of total energy consumption.

Energy inputs for crop management

The potato crop was propagated through tubers; garden pea with seeds; and tomato, muskmelon, and cauliflower seedlings were transplanted. Hence, potato crop required higher seed rate (3,250–4,500 kg.ha^{-1}) compared to garden pea (75–112.5 kg.ha^{-1}), muskmelon (1.0 kg.ha^{-1}), cauliflower (0.625–1.250 kg.ha^{-1}), and tomato (0.250–0.300 kg.ha^{-1}), Table 2. Consequently, potato being a vegetatively propagated crop consumed a significantly ($p<0.05$) higher seed energy (14,100 ± 3,523 MJ.ha^{-1}) than other crops, specifically garden pea (194.8 ± 45.8 MJ.ha^{-1}), cauliflower (1.1 ± 0.4 MJ.ha^{-1}), muskmelon (0.8 MJ.ha^{-1}), and tomato (0.2 MJ.ha^{-1}).

The energy contributed by FYM was significantly

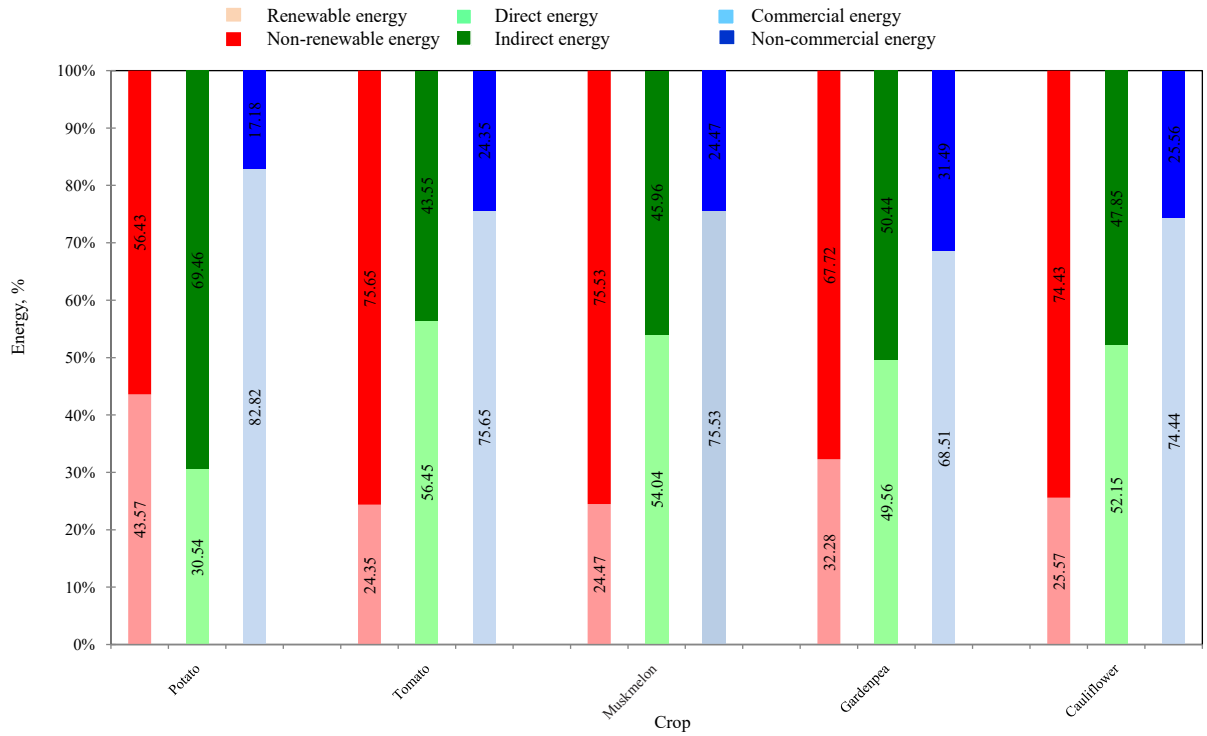

Fig. 3: Categorization of share of various sources of energy in production of selected vegetable crops

higher ($p<0.05$) in production of potato, tomato, and cauliflower (8,250± 750 MJ.ha^{-1} for each crop) than muskmelon (7,500± 300 MJ.ha^{-1}) and garden pea (6,150± 150 MJ.ha^{-1}). The nitrogen requirement was higher for potato crop (187.50–200 kg.ha^{-1}) than tomato (150–155 kg.ha^{-1}), cauliflower (125–140 kg.ha^{-1}), muskmelon (120–125 kg.ha^{-1}), and garden pea (50–75 kg.ha^{-1}) crops. The consequential significant ($p<0.05$) consumption of fertiliser energy was determined to be higher in potato (12,742 ± 425 MJ.ha^{-1}) and tomato (10,404 ± 214 MJ.ha^{-1}) than cauliflower (9,109 ± 500 MJ.ha^{-1}), and muskmelon (8,471 ± 226 MJ.ha^{-1}). However, garden pea (4,647 ± 843 MJ.ha^{-1}) crop consumed lowest amount of fertiliser energy at 5% level of significance (Table 5).

Pesticide use in different vegetable crops depends on the incidence of disease, insects, and pests. The study found that the highest pesticide consumption was for potato and tomato crops. No herbicide was used for muskmelon and cauliflower crops, while pre-emergence herbicides were used in tomato and occasionally in potato for controlling weeds. Yet, post-emergence herbicides were used mainly in potato crop production. Hence, a significantly ($p<0.05$) higher chemical energy was used in potato (1,040 ± 121 MJ.ha^{-1}) and tomato (870 ± 120 MJ.ha^{-1}) crop. However, there was an insignificant ($p<0.05$) difference in use of chemical energy (600 ± 60 - 660 ± 180 MJ.ha^{-1}) for the other crops. This was due to higher chemical application rate for potato (6.25–8.25 kg.ha^{-1}) and tomato crop (7.50–9.50 kg.ha^{-1}); whereas the application rate was nearly similar for muskmelon (4.0–6.0 kg.ha^{-1}), garden pea (4.5–5.5 kg.ha^{-1}), and cauliflower (4.0–7.0 kg.ha^{-1}).

Electric energy

Among all energy sources, the contribution of electricity, exclusively used in irrigation operation, was high for cultivation of the selected crops (Fig. 4). Electricity was used for operating submersible pumps for irrigation. Significantly higher ($p<0.05$) electric energy was consumed in the production of tomato (13,803 ± 584 MJ.ha^{-1}), followed by muskmelon (11,850 ±645 MJ.ha^{-1}), cauliflower (11,513 ± 1,729 MJ.ha^{-1}), potato (5,289± 315 MJ.ha^{-1}), and garden pea (2,830 ± 51.5 MJ.ha^{-1}), Table 5.

Human energy

Most of the farm operations as pre-planting seed sorting/drying/treatment, transplanting of cauliflower/ tomato seedling, manual spraying, harvesting/plucking of plant/ fruit (except potato), and loading/unloading of produce in tractor trailer were carried out by human labour only (Figs. 5, 6). Significantly higher ($p<0.05$)

Table 5. Energy inputs from various sources for production of selected vegetable crops

SI. No.	Energy source	Farm operation	Energy input, MJ.ha^{-1} Potato			Tomato			Muskmelon			Garden pea			Cauliflower		
1.	Human	Seed drying/ treatment	$^{b}189.8^{A}$	±	13.9	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Nursery raising	0.0	±	0.0	$^{b}142.2^{A}$	±	3.2	$^{b}148.9^{A}$	±	3.2	$^{b}7.1^{B}$	±	0.7	$^{bc}143.5^{A}$	±	3.0
		Tillage	$^{ef}15.8^{AB}$	±	1.5	$^{b}18.3^{A}$	±	1.3	$^{b}18.6^{A}$	±	1.3	$^{b}14.1^{B}$	±	1.2	$^{d}16.9^{AB}$	±	1.3
		Sowing/planting	$^{d}49.0^{A}$	±	9.8	0.0	±	0.0	0.0	±	0.0	$^{b}7.9^{B}$	±	0.8	0.0	±	0.0
		Transplanting	0.0	±	0.0	$^{b}136.9^{A}$	±	5.4	$^{b}140.5^{A}$	±	5.4	0.0	±	0.0	$^{bc}147.2^{A}$	±	6.2
		Bund making	$^{f}1.0^{A}$	±	0.0	$^{b}0.9^{A}$	±	0.1	$^{b}1.0^{A}$	±	0.1	$^{b}1.0^{A}$	±	0.1	$^{d}1.0^{A}$	±	0.1
		Interculture	$^{de}36.0^{B}$	±	2.4	$^{b}33.2^{B}$	±	2.2	$^{b}32.8^{B}$	±	2.2	$^{b}21.3^{B}$	±	18.5	$^{bc}138.3^{A}$	±	16.1
		Transportation/ application of FYM/ Fertiliser	$^{ef}19.7^{A}$	±	2.0	$^{b}18.7^{A}$	±	0.9	$^{b}18.4^{A}$	±	0.9	$^{b}17.9^{A}$	±	1.7	$^{d}19.0^{A}$	±	1.0
		Spraying	$^{f}5.5^{C}$	±	0.3	$^{b}130.1^{A}$	±	7.4	$^{b}86.2^{B}$	±	7.4	$^{b}64.2^{B}$	±	14.7	$^{cd}80.3^{B}$	±	8.0
		Irrigation	$^{c}93.2^{B}$	±	5.6	$^{b}243.2^{A}$	±	10.3	$^{b}208.8^{A}$	±	10.3	$^{b}47.6^{C}$	±	5.5	$^{b}202.8^{A}$	±	30.4
		Harvesting/ digging/plucking	$^{a}490.2^{C}$	±	20.3	$^{a}2475.9^{A}$	±	311.9	$^{a}1052.6^{B}$	±	311.9	$^{a}1361.1^{B}$	±	94.3	$^{a}1018.7^{B}$	±	88.5
		Transportation of produce	$^{def}23.4^{B}$	±	6.8	$^{b}112.8^{A}$	±	32.3	$^{b}47.3^{B}$	±	32.3	$^{b}61.5^{AB}$	±	19.8	$^{d}46.2^{C}$	±	12.9
		Sub Total	923.7^{C}	±	19.2	3312.3^{A}	±	236.2	1755.0^{B}	±	228.2	1603.8^{B}	±	89.7	1813.9^{B}	±	101.7
2.	Machinery	Seed drying/ treatment	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Nursery raising	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Tillage	$^{b}108.6^{B}$	±	8.3	$^{c}127.9^{AB}$	±	6.1	$^{c}130.3^{AB}$	±	9.0	$^{b}110.7^{AB}$	±	8.7	$^{c}132.3^{A}$	±	9.8
		Sowing/planting	$^{b}143.5^{A}$	±	28.7	0.0	±	0.0	0.0	±	0.0	$^{c}42.2^{B}$	±	4.3	0.0	±	0.0
		Transplanting	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Bund making	$^{d}7.8^{A}$	±	0.2	$^{e}7.5^{A}$	±	0.3	$^{e}7.7^{A}$	±	0.3	$^{c}7.7^{A}$	±	0.5	$^{e}7.7^{A}$	±	0.5
		Interculture	$^{cd}25.9^{A}$	±	1.8	$^{e}4.3^{B}$	±	0.3	$^{e}4.2^{BC}$	±	0.4	$^{c}0.3^{C}$	±	0.3	$^{e}2.0^{C}$	±	0.0
		Transportation/ application of FYM/ Fertiliser	$^{a}373.2^{A}$	±	37.5	$^{e}353.0^{A}$	±	16.9	$^{a}348.2^{A}$	±	19.5	$^{a}338.9^{A}$	±	30.9	$^{a}359.5^{A}$	±	19.5
		Spraying	$^{c}61.8^{A}$	±	3.1	$^{e}20.8^{A}$	±	1.2	$^{e}13.8^{C}$	±	2.0	$^{c}10.3^{C}$	±	2.4	$^{e}12.8^{C}$	±	1.3
		Irrigation	$^{cd}36.1^{B}$	±	2.2	$^{d}94.1^{A}$	±	4.0	$^{d}80.8^{A}$	±	4.4	$^{c}19.1^{C}$	±	2.9	$^{d}78.5^{A}$	±	18.0
		Harvesting/ digging/plucking	$^{c}60.5^{A}$	±	7.3	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Transportation of produce	$^{b}151.3^{C}$	±	15.5	$^{a}550.0^{A}$	±	29.8	$^{b}230.3^{B}$	±	25.9	$^{a}300.7^{B}$	±	41.3	$^{b}226.0^{BC}$	±	26.6
		Sub Total	968.7^{B}	±	81.5	$1157.6.2^{A}$	±	31.4	815.4^{C}	±	44.6	830.0^{BC}	±	61.7	818.8^{C}	±	41.2
3.	Diesel	Seed drying/ treatment	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Nursery raising	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Tillage	$^{b}1738.4^{B}$	±	131.9	$^{c}1843.9^{AB}$	±	94.7	$^{c}1877.9^{AB}$	±	132.5	$^{b}1762.7^{B}$	±	138.3	$^{b}2194.5^{A}$	±	174.7
		Sowing/planting	$^{b}2053.0^{A}$	±	410.6	0.0	±	0.0	0.0	±	0.0	$^{c}496.0^{B}$	±	49.9	0.0	±	0.0
		Transplanting	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Bund making	$^{e}123.2^{A}$	±	3.1	$^{d}118.4^{A}$	±	4.4	$^{d}122.3^{A}$	±	4.6	$^{cd}122.5^{A}$	±	7.6	$^{c}121.4^{A}$	±	7.2
		Interculture	$^{de}353.1^{A}$	±	25.3	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
		Transportation/ application of FYM/ Fertiliser	$^{a}2976.2^{A}$	±	299.1	$^{b}2815.3^{A}$	±	134.9	$^{a}2777.1^{B}$	±	155.2	$^{a}2702.5^{A}$	±	246.7	$^{a}2866.7^{A}$	±	155.2
4		Spraying	$^{cd}695.1^{A}$	±	34.6	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
5.		Irrigation	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
6.		Harvesting/ digging/plucking	$^{c}968.4^{A}$	±	116.1	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0	0.0	±	0.0
7.		Transportation of produce	$^{c}1190.5^{C}$	±	119.6	$^{a}4914.3^{A}$	±	266.0	$^{b}2057.9^{B}$	±	231.9	$^{a}2686.4^{B}$	±	369.4	$^{b}2019.3^{B}$	±	237.2
		Sub Total	10097.9^{A}	±	945.5	9691.9^{A}	±	366.8	6835.2^{B}	±	392.7	7770.1^{B}	±	580.8	7201.9^{B}	±	393.7
8.	Electricity	Irrigation	5289.3^{B}	±	315.3	13803^{A}	±	584	11849.5^{A}	±	645.1	2830.1^{C}	±	451.5	11513.4^{A}	±	1728.6
9.	Seed	Input for crop management	14100^{A}	±	3523.3	0.2^{B}	±	0.00	0.8^{B}	±	0.0	194.8^{B}	±	45.8	$^{d}1.1^{B}$	±	0.4
10.	FYM	Input for crop management	8250.0^{A}	±	750.0	8250.0^{A}	±	750.0	7500.0^{AB}	±	300.0	6150.0^{B}	±	150.0	8250.0^{A}	±	750.0
11.	Fertiliser	Inputs for crop management	12742.3^{A}	±	425.1	10404.5^{A}	±	214.6	8470.7^{C}	±	226.3	4646.6^{D}	±	843.4	9107.8^{BC}	±	500.8
12.	Chemical	Inputs for crop management	1040.0^{A}	±	121.2	870.0^{AB}	±	120.0	600.0^{B}	±	120.0	600.0^{B}	±	60.0	660.0^{B}	±	180.0
13.	Total energy input		$53,412^{A}$	±	2388	$47,489^{B}$	±	1183	$37,827^{C}$	±	856	$24,625^{D}$	±	497	$39,367^{C}$	±	1127

Note: *1. Values are expressed as Mean± Standard deviation*
2. Difference superscripts are significantly different ($p<0.05$) using Tukey's test
3. Small notations give comparison within the energy sources used under specified farm operations in rows
4. Capital notations gives comparison between the vegetable crops in columns

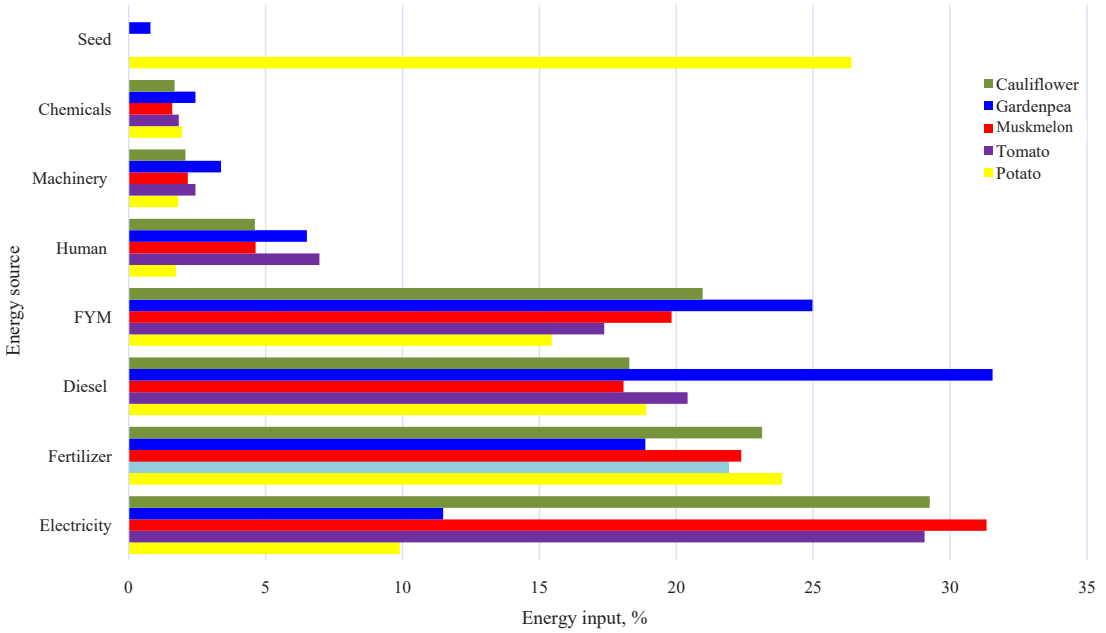

Fig. 4: The contribution of energy consumption from various sources of energy for production of selected vegetable crops

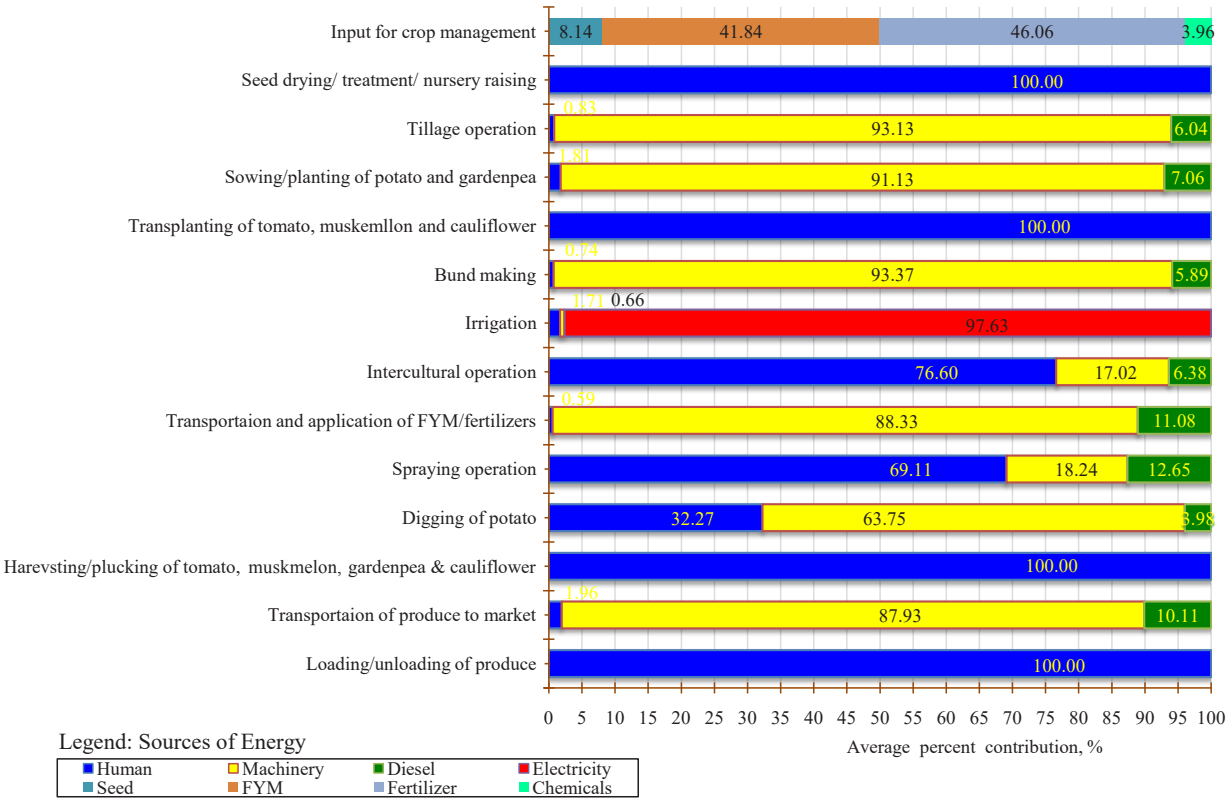

Fig. 5: Average percent contribution of various energy sources in farm operations of vegetable crop production

(i) Potato seed sorting before planting (ii) Cauliflower nursery (iii) Tomato seedling transplanting

(iv) Irrigation (v) Fertiliser distribution (vi) Manual knapsack spraying (vii) Muskmelon harvesting

(viii) Garden pea plucking (ix) Pototo collection/ bagging (x) Loading/ unloading of produce in tractor-trailer

Fig. 6: Farm operations using human energy for vegetable crop production

human energy (3,312 ± 236.2 MJ.ha^{-1}) was consumed in the production of tomato crop than the other crops. The energy consumed in cauliflower production (1,813.9 ± 101.7 MJ.ha^{-1}), muskmelon (1,755 ± 228.2 MJ.ha^{-1}), and garden pea (1,603.8 ± 89.7 MJ.ha^{-1}) were statistically equivalent (Table 5). Some operations in potato production were mechanised, and thus a significantly lower human energy (923.7± 19.2 MJ.ha^{-1}) was required. Higher human energy consumption in tomato cultivation was mainly due to higher involvement of human energy for raising seedling nursery and transplanting, and multiple manual fruit plucking.

Energy from machinery and diesel

Although majority of the farm operations were performed through machines for potato production as compared to the other vegetable crops (Fig. 7), tomato had significantly higher ($p<0.05$) machinery energy consumption (1157.6 ± 31.4 MJ.ha^{-1}). This was because of multiple plucking (manual) and subsequent multiple transportations of produce to the market as tomato is a highly perishable crop.

Tomato crop needed five times of manual plucking, whereas garden pea required 2-3 plucking, while

(i) Tillage operation (ii) Potato planting (iii) Garden pea sowing

(iv) Bund making (v) Soil earthing up (vi) Spraying

(vii) Transportation of FYM to the field (viii) Potato digging (ix) Transportation of produce to market

Fig. 7: Energy from machinery and diesel for various farm operation in vegetable crops production

muskmelon and cauliflower crops needed only two plucking (Table 2). The energy contributed by machines for the other crops was 968.7 ± 81.5 MJ.ha^{-1} for potato, 818.8 ± 41.2 MJ.ha^{-1} for cauliflower, 815.4± 44.6 MJ.ha^{-1} for muskmelon, and 830.0 ± 61.7 MJ.ha^{-1} for garden pea crop.

Diesel energy was consumed as fuel for operating farm machines and equipment using tractor. As presented in Table 5, the total diesel energy consumption was significantly higher ($p<0.05$) in potato (10,098 ± 945 MJ.ha^{-1}) than tomato (9,692 ± 367 MJ.ha^{-1}), muskmelon (6,835 ± 393 MJ.ha^{-1}), garden pea (7,770 ± 581 MJ.ha^{-1}), and cauliflower (7,202 ± 394 MJ.ha^{-1}).

Energy Consumption in Various Farm Operations

Figure 8 represents the percent contribution of energy consumed out of the total energy input in various farm operations. The data from the figure signified that 41.11 - 67.64% of total input energy was contributed by various inputs for crop management, specifically seed, FYM, fertilisers, and chemicals. Among the farm operations, irrigation consumed highest energy (10.14–32.09%), followed by transportation and application of FYM/fertilisers (6.31–12.42%), and transportation of produce (2.56–12.38%). The contribution of tillage operation for cultivation of the crops was in the range of 3.49–7.66% of total energy input. The average percent of energy consumed in sowing/planting/transplanting operation of potato and cauliflower crops was 4.20% and 2.22%, respectively, to the total operational energy; whereas it was less than 0.5% in the other crops. The results also showed that the energy contribution for spraying, seed drying/nursery raising, intercultural

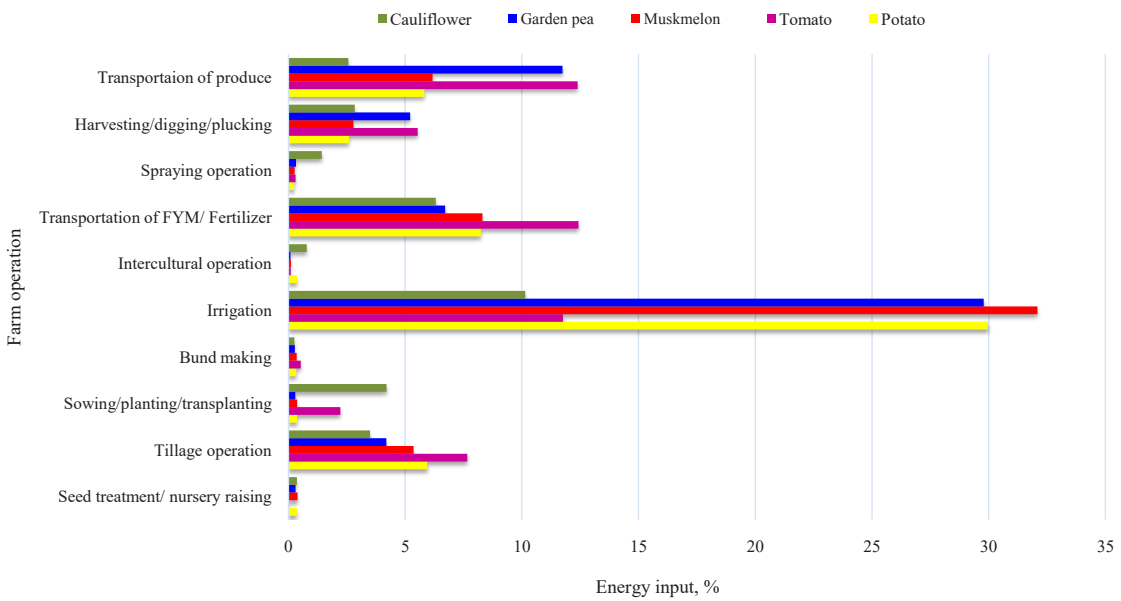

Fig. 8: Requirement of energy consumption in various farm operations for production of selected vegetable crops

operation, and bund making in each crop was less than 1.0%, except in spraying operation of potato crop (1.43%).

During tillage operation, no significant difference ($p<0.05$) in energy consumption was observed among the selected crops due to almost similar cultivation practices. In Punjab agriculture, planting of potato seed was done by automatic/semi-automatic potato planter, while tractor-operated ridger-seeder or pea seed-drill was used for sowing of garden pea seed. Conversely, transplanting operations of tomato, muskmelon, and cauliflower seedlings were performed manually consuming only human energy (100%) for preparation of nursery as well as transplanting of these crops. After sowing/planting/transplanting operation, bunds were formed for irrigation with tractor-operated bund maker. The energy consumed for bund making for crops was similar (126.8 - 132.0 MJ.ha^{-1}) because of the requirement of similar plots for irrigation, and was contributed by 93.37 % machinery, 5.89% diesel, and 0.74% human energy (Fig. 3).

The results indicated about 97.63% of average energy consumed in irrigation operation was utilized from electric energy, whereas the energy from human and machinery sources was 1.71% and 0.66%, respectively, for the selected crops (Fig. 3). Considering the total energy consumed for irrigation, significantly higher ($p<0.05$) energy was consumed in tomato (14,140 ± 598 MJ.ha^{-1}) and muskmelon (12,139 ± 661 MJ.ha^{-1}) crops, followed by cauliflower (11,795 ± 1,771 MJ.ha^{-1}), potato (5,419 ± 138 MJ.ha^{-1}), and garden pea (2,897 ± 460 MJ.ha^{-1}). The variation in consumption of electric energy among these crops was because of variation in the quantity of irrigation requirement for each crop (Table 2). A wide distribution range of irrigation energy for different crops is attributed to the fact that vegetables are grown during different seasons, and hence have diverse irrigation requirements. Tomato, muskmelon, and cauliflower are grown during summer, and require higher frequency of irrigation, resulting in more energy consumption. No significant difference ($p<0.05$) was observed between energy consumption for transporting and spreading of FYM/fertilisers for all crops because of adoption of similar techniques/ methods for transporting and spreading of FYM/ fertilisers in the field.

Total energy consumed for interculture operation (weeding/ hoeing/ earthing-up) was significantly ($p<0.05$) higher in potato crop (415 ± 30 MJ.ha^{-1}), followed by cauliflower (140.3 ± 16.1 MJ.ha^{-1}), tomato (37.5 ± 2.5 MJ.ha^{-1}), muskmelon (37.0 ± 3.1 MJ.ha^{-1}), and garden pea (21.7 ± 18.8 MJ.ha^{-1}) crops. Higher energy consumption in potato crop might be due to additional energy from machinery and diesel for earthing-up of soil. Variation in energy consumption in cauliflower crop was due to multiple weeding operations (4 times more than the other crops).

Total energy consumed in harvesting/digging/plucking

of produces was significantly higher ($p<0.05$) in tomato crop (2,476 ± 375 MJ.ha^{-1}) than the other crops. However, there is no significant difference in harvesting energy consumption between the other crops (1,052 ± 227 – 1,361 ± 94.3 MJ.ha^{-1}). Tomato crop required five times energy for manual plucking; whereas garden pea crop required 2-3 times, and muskmelon / cauliflower crop required two times (Table 2). Harvesting of prime produce from tomato, muskmelon, garden pea, and cauliflower crops was performed by human labour. Therefore, the energy consumed for harvesting/ plucking operation of these crops was fully contributed by human energy (Fig. 3). In contrast, potato being a root crop, used tractor-operated potato digger for digging the tubers. After mechanical digging, potato tubers were manually collected from the field and bagged.

Transportation operations were performed for loading/ unloading and carrying the final produces to the market through tractor trailer after every harvesting/plucking/ picking operation. Multiple harvesting/plucking were required for most of the selected crops; therefore, transportation operations varied with the number of times of harvesting. Significantly, a higher contribution ($p<0.05$) of total energy was observed in transportation operation of tomato (5,577 ± 270 MJ.ha^{-1}), followed by garden pea (3,049 ±409 MJ.ha^{-1}), muskmelon (2,335 ± 258 MJ.ha^{-1}), cauliflower (2,291 ± 262 MJ.ha^{-1}), and potato crop (1,365 ± 134 MJ.ha^{-1}). Higher energy consumption in transportation operation was observed in tomato crop because of its higher yield and higher frequency of harvesting than the other crops.

Energy Consumption Analysis

Input-output energy

Among different energy sources, the total energy inputs differed for every crop production. A significantly higher ($p<0.05$) energy consumption was observed in potato (53,412 ± 2,388 MJ.ha^{-1}) cultivation than tomato (47,489 ± 1,183 MJ.ha^{-1}), cauliflower (39,367 ± 1,127 MJ.ha^{-1}), muskmelon (37,827 ± 856 MJ.ha^{-1}), and garden pea crop (24,625 ± 497 MJ.ha^{-1}).

Among the selected vegetable crops, the energy output was significantly different from each other at 5% level of significance (Table 6). Highest energy output was observed in potato crop (104,400 ± 3,600 MJ.ha^{-1}), followed by tomato (48,800 ± 800 MJ.ha^{-1}), cauliflower (41,400 ± 600 MJ.ha^{-1}), muskmelon (23,200 ± 800 MJ.ha^{-1}), and garden pea (13,775 ± 475 MJ.ha^{-1}).

Energy-use efficiency

Energy-use efficiency (EUE) was significantly higher ($p<0.05$) for potato (1.96). However, EUE of cauliflower and tomato crops were 1.05 and 1.03, respectively. The input energies of muskmelon and garden pea crops were higher than the output energy; therefore, the net energy was negative and the EUE was 0.61 and 0.56, respectively. The cultivation of a crop would be profitable in terms of energy consumed when the EUE would be greater than 1.0. The findings of some of the previous researchers showed that the EUE was 2.74 for tomato crops (Sepat *et al.,*2013), 0.52 (Kuswardhani *et al.,* 2013), and 1.78 (Bilalis *et al.,* 2013) under open field conditions. Higher EUE value of 8.9 in potato crop was also reported (Lin *et al*., 2017). EUC less than 1.0 was obtained in garden pea crop (Raghuvanshi *et al.,* 2018). However, EUE of muskmelon crop production was 1.9. The energy-use efficiency varied from crop to crop and even from location to location.

Potato and garden pea crops had low consumptions of human energy as crop production were predominantly mechanised. Agricultural operations for potato crop were highly mechanised, leading to higher diesel consumption. Garden pea sowing was also mechanised; except for harvesting, but the other three crops had less mechanised operations apart from tillage, bund making, fertiliser application and transportation. Thus, there was a large difference in diesel energy consumption for potato and the other crops. Diesel consumption was higher for produce transportation in tomato due to its higher yield volume, frequent plucking, careful handling requirement during transportation because of its high perishability, and also immediate transportation from the field to the market involving higher distance travels. The results also showed that tractor-trailers consumed maximum energy for vegetable crop production. Although, the total energy input was lesser in muskmelon crop, the energy consumption for irrigation operation was significantly higher than potato and cauliflower.

The above discussion indicate that the input and output energy involved in the production of a crop depend on several factors, e.g., topography of the area, growing season, and growing conditions (irrigated or rainfed, open field or greenhouse). Due to significant contribution of non-renewable and commercial energy in the production of different crops, the agricultural system is vulnerable to increase in energy price through

Table 6. Energy indices in major vegetable crop production

Sl. No	Energy parameter	Potato	Tomato	Muskmelon	Garden pea	Cauliflower
1.	Crop yield, $kg.ha^{-1}$	$29,000^{C} \pm 1000$	$61,000^{A} \pm 1000$	$29,000^{C} \pm 1000$	$7,250^{D} \pm 250$	$34,500^{B} \pm 500$
2.	Output energy, $MJ.ha^{-1}$	$104,400^{A} \pm 3600$	$48,800^{B} \pm 800$	$23,200^{D} \pm 800$	$13,775^{E} \pm 475$	$41,400^{C} \pm 600$
3.	Total energy input, $MJ.ha^{-1}$	$53,412^{A} \pm 2389$	$47,489^{B} \pm 1183$	$37,827^{C} \pm 856$	$24,625^{D} \pm 497$	$39,367^{C} \pm 1127$
4.	Net energy, $MJ.ha^{-1}$	$50,988^{A} \pm 1521$	$1,311^{B} \pm 459$	$-14627^{D} \pm 1319$	$-10850^{C} \pm 850$	$2,033^{B} \pm 1169$
5.	Energy-use efficiency	$1.96^{A} \pm 0.04$	$1.03^{B} \pm 0.01$	$0.61^{C} \pm 0.03$	$0.56^{C} \pm 0.03$	$1.05^{B} \pm 0.03$
6.	Specific energy consumption, $MJ.kg^{-1}$	$1.84^{B} \pm 0.03$	$0.80^{D} \pm 0.01$	$1.26^{C} \pm 0.06$	$3.40^{A} \pm 0.16$	$1.14^{C} \pm 0.04$
7.	Energy productivity, $kg.MJ^{-1}$	$0.54^{D} \pm 0.02$	$1.29^{A} \pm 0.02$	$0.77^{C} \pm 0.03$	$0.29^{E} \pm 0.02$	$0.88^{B} \pm 0.02$

Note: 1. Values are expressed as Mean± Standard deviation
2. The difference superscripts are significantly different ($p<0.05$) using Tukey's test
3. Capital notations give comparison between the vegetable crops

petroleum as well as fertiliser prices. Moreover, the energy-use pattern, energy-use efficiency, specific energy consumption, and other related parameters vary according to the type of crop and agro-climatic region; therefore, there is a need to involve more energy-efficient farm machinery and irrigation systems. Fertiliser consumption can also be minimized by utilizing other techniques as use of crop residue to substitute nutrient supply.

CONCLUSIONS

To determine the effects of energy input levels on vegetable crop production in Hoshiarpur, Jalandhar, and SBS Nagar districts of Punjab, the energy-use analysis (both source and farm operation) of five major vegetable production systems were investigated. Significantly higher ($p<0.05$) energy consumption was observed in potato (53,412 ± 2,388 $MJ.ha^{-1}$) production than tomato (47,489 ± 1,183 $MJ.ha^{-1}$), cauliflower (39,367 ± 1,127 $MJ.ha^{-1}$), muskmelon (37,827 ± 856 $MJ.ha^{-1}$), and garden pea (24,625 ± 497 $MJ.ha^{-1}$). Significantly higher ($p<0.05$) energy (3.40 $MJ.kg^{-1}$) was used to produce one kg seed of garden pea crop, followed by potato (1.84 $MJ.kg^{-1}$), muskmelon (1.26 $MJ.kg^{-1}$), and cauliflower crops (1.14 $MJ.kg^{-1}$). Lowest specific energy (0.80 $MJ.kg^{-1}$) was required for production of tomato. Irrigation was the major farm operation that consumed highest energy, followed by transportation and application of FYM/fertilisers, transportation of produce, tillage, and harvesting/plucking operation.

The average contributions of total renewable sources were in the range of 25.57 - 43.57% of total input energy in the selected vegetable crops. Non-renewable energy sources contributed about 56.43 - 74.43% of total energy input for production of the vegetable crops. Direct and indirect energy sources contributed 30.54 - 54.04% and 47.85 - 69.46%, respectively, of total energy; whereas commercial and non-commercial energy sources contributed 68.51 - 82.82% and 17.18 - 31.5%, respectively, of total energy. A need, therefore, exists for more energy-efficient farm machinery and irrigation systems. Rationalised fertiliser consumption with supplementation with crop residue can reduce the share of commercial energy uses.

CONFLICTS OF INTEREST

The authors declare that there is no conflict of interest in any form that could have influenced the research work reported in this paper. Competing interest in disclosure of the research work has been considered by the authors.

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